Spinal X-ray
Shared HRNet
Stage 1
Stage 2
(b) ECFM
F_ℓ
Stage 3
(b) ECFM
F_ℓ
Stage 4
(b) ECFM
F_ℓ
Heatmap head
H^0
Initial prediction
Interactive correction
(a) Click & drag
4
C^{ic}
H^{ic}
(c) RJC
Corrected prediction

(a) Interactive correction

Drag to correct ④
Initial position
Interaction point
Interaction information
C^{ic} …
Interaction context
$F(o_4)$
$F(q_4)$
δ_4
d_4^A
a_4^{ctx}
Cross-attention
Q_8
K_4
A_{84}
V_4
c_8^A

(b) Evidence-conditioned Feature Modulation (ECFM)

Evidence re-localization
Feature modulation
Dual-branch evidence sampling
Local
$f_{8m}^{L,\ell}$
r_8
L
Structural
$f_{8m}^{S,\ell}$
s_8, B_8
S
Target
f_8^{tar}
Context
c_8^A
Evidence
e_8^{fuse}
Landmark embedding
t_8
Channel modulation
F_ℓ
$1+\varepsilon \tanh(\gamma_\ell)$
$\varepsilon\kappa_\ell$
Spatial modulation
$W_s^\ell t_8$
$\hat{M}_8^\ell$
$\varepsilon\sum_i$
F_ℓ^+
Evidence matching
ψ_8^ℓ
Compatibility
Weighted evidence
α_8^L
α_8^S
e_8^L
e_8^S
g
e_8^{fuse}
Heatmap modulation
H^0
F^+
Shared heatmap head
H^{ic}

(c) Relation-guided Joint Correction (RJC)

Joint correction
Relation prediction
t_4
t_8
f_R, f_Δ
Pairwise geometry
$\xi_4 = q_4 - \hat{p}_4^0$
ω_{48}, d_{48}
Constrained joint inference
$\Lambda^0 + B^T W B$
$\hat{\Xi}^{ic}$
Visual displacement
H^0 μ^0
H^{ic} μ^{ic}
v
$v = \mu^{ic} - \mu^0$
Relations on initial estimates
ω_{68}
d_{48}
ω_{48}
$\hat{p}_4^{ic} = q_4$
Visual → joint correction
Initial
Visual
Joint
v_i
ω_{ij}
d_{ij}
Corrected landmarks
$\hat{P}^{ic} = \hat{P}^0 + \hat{\Xi}^{ic}$